

Flip-chip Integrated Differential Drive EML with Dual Modulator Structure with AC-coupling for 226 Gb/s

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Abstract We report a flip-chip-mounted EML with a dual-EAM structure on an interposer, together with differential termination and an AC-coupling capacitor. The device exhibits >60 GHz 3-dB electro-optic bandwidth and 226 Gb/s back-to-back operation at 113-GBaud PAM4 with 2.8-dB TDECQ. ©2026 The Author(s)

Introduction

Advanced AI systems based on large language models (LLM) demand enormous compute and very high-bandwidth interconnections. For AI clusters and data centre optical interconnects, increasing capacity requires both higher bit rates per lane and a denser transmitter package. Silicon photonic ring modulators are highly attractive for compact, high-density integration [1], while thin-film lithium niobate (TFLN) modulators remain promising candidates for achieving the ultra-wide bandwidths required for 400 Gbps/lane operation [2]. Electro-absorption modulator integrated laser diodes (EMLs), which monolithically integrate a light source and a compact, high-speed modulator on a single chip, are widely used in today's data centre optical interconnects and remain strong candidates for next-generation high-density transmitters [3–6]. Efforts toward higher integration density include EML arrays [7], shortened wire lengths [8], and flip-chip mounting [9]. Furthermore, recent demonstrations of membrane III-V EAMs integrated with differential CMOS drivers highlight the growing importance of tighter photonic-electronic co-integration for high-density transmitters [10].

EMLs are conventionally mounted on a submount and interconnected by wire bonding. Moreover, passive components such as termination resistors and AC-coupling capacitors are required around the device. In particular, differential-drive EMLs require more such passive elements, which increases the footprint of each channel and limits the achievable integration density. In this work, we propose and demonstrate a flip-chip-mounted EML featuring a differential-drive-capable dual-EAM structure, together with 100-Ω terminations and AC-coupling capacitors integrated on the package substrate. The device exhibits a differential 3-dB electro-optic (EO) bandwidth exceeding 60 GHz

and enables back-to-back (B2B) optical operation at 226 Gb/s (113-GBaud PAM4). To the best of our knowledge, this is the first reported flip-chip bonding of an EML employing a dual-EAM structure with both integrated terminations and AC coupling on the package substrate. These results demonstrate a practical route toward higher integration density in multichannel transmitter modules.

Device structure and concept

Figure 1 shows a schematic of the differential-drive-capable EML chip used in this study. The single die integrates a buried-heterostructure (BH) distributed-feedback laser diode (DFB-LD), a high-mesa electro-absorption modulator (EAM), and a spot-size converter (SSC). The BH structure enables high-output-power DFB operation, while the high-mesa EAM provides low static capacitance and strong optical confinement for high-speed operation. In general, differential drive with a 100-Ω termination reduces the effective bandwidth compared with single-ended drive (Fig. 2(a)), often by approximately a factor of two.

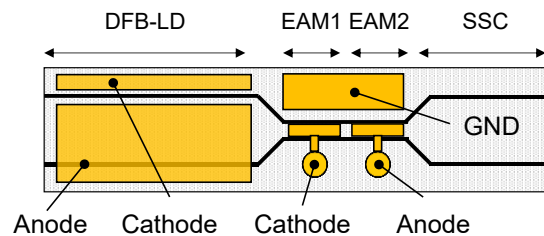


Fig. 1: Schematic structure of the EML chip

To mitigate this limitation, our device employs two EAM sections connected in series (series dual EAM, Fig. 2(b)) [3]. For a given total modulator length, this series partitioning reduces the capacitance of each section compared to a single EAM

of the same length, thereby enabling faster differential operation. The two EAM sections share a common ground while allowing independent biasing of each section.

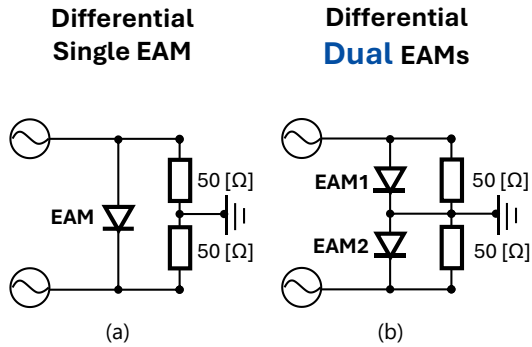


Fig. 2: Schematic diagram of differential EMLs. (a) Differential EML for anode-cathode drive, and (b) differential EML with tandem dual EAMs

Module configuration

Figure 3 shows the prototype module. GSGSG transmission lines for RF probing are formed on the package substrate and then converted to GSSG lines for differential signal delivery. The differential lines on the package substrate are designed with a differential impedance of 100 Ω . The differential signals are delivered to the EAM through TGVs in the 120- μm -thick glass interposer (Glass IP), providing a short electrical path to the modulator. The LD electrodes are likewise connected to the package-substrate wiring through vias in the Glass IP. The package substrate, with a thickness of 230 μm , the Glass IP, and the EML are assembled using Au stud bumps. Differential termination resistors are integrated in the Glass IP using thin-film tantalum nitride and are designed as 50- Ω resistors on each side, corresponding to a 100- Ω differential termination. For AC coupling, 10-nF 0201M (0.2 mm $\times$ 0.1 mm) chip capacitors are mounted on the backside of the substrate, thereby preventing DC current from flowing through the termination resistors. In addition to avoiding the parasitic inductance associated with wire bonds, the flip-chip configuration is expected to reduce manufacturing variations such as wire-length nonuniformity.

Experimental result

All measurements were performed with the package temperature set to 55 $^{\circ}\text{C}$ using a thermoelectric cooler (TEC). Under the bias conditions shown in Fig. 4, the DFB operated at a lasing wavelength of 1316 nm with a side-mode suppression ratio (SMSR) greater than 40 dB.

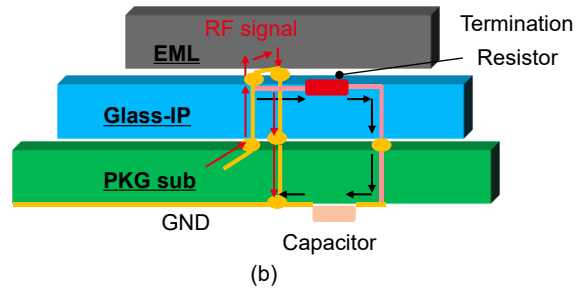
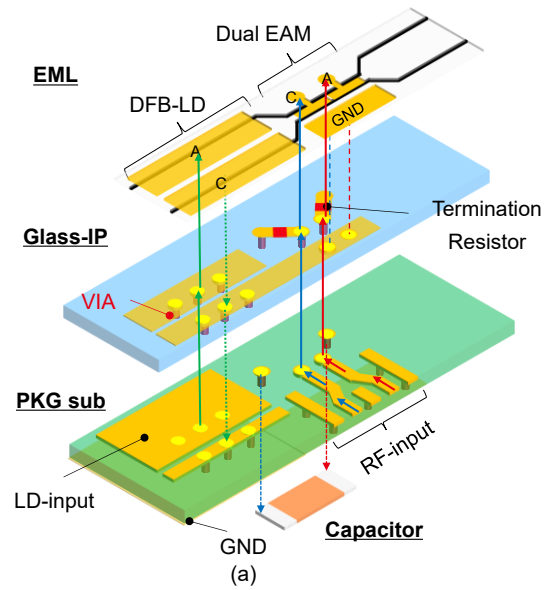


Fig. 3: A fabricated prototype of flip-chip bonded EML with glass-IP on the package substrate (a) Perspective view, (b) Side view

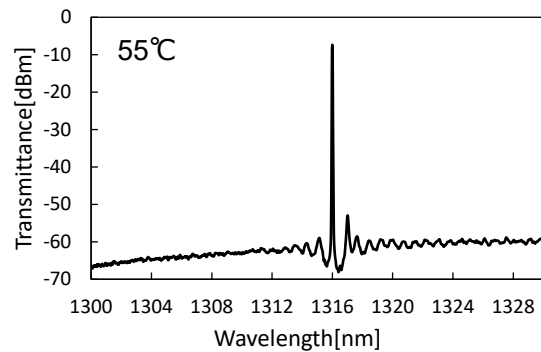


Fig. 4: Measured lasing spectrum at 55 $^{\circ}\text{C}$

Figure 5 shows the measured electro-optic (EO) S21 response. Measured with a network analyser and RF probes connected to the EML, the differential 3-dB EO bandwidth is 60.3 GHz. The peaking observed around 30 GHz is attributed to the vias formed in the Glass IP, which are effectively inserted in series with the termination resistors and provide a response like that of the peaking inductors used in previously reported EML sub-assemblies [9].

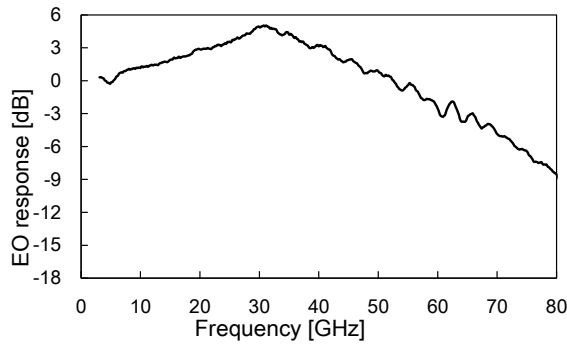


Fig. 5: Schematic structure of the EML chip

For PAM4 characterization, Figure 6 shows the measured optical waveform of the 226 Gb/s PAM4 signal. The 113.4375-Gbaud SSPRQ test pattern was generated by a 256-GSa/s arbitrary waveform generator (AWG). The DFB drive current was 40 mA, and the electrical drive swing was approximately 1.0 Vpp. The optical output was filtered by a fourth-order Bessel–Thomson filter with a 56.72-GHz bandwidth before detection. At the receiver, a 15-tap feed-forward equalizer (FFE) was used for TDECQ evaluation with a target symbol error rate (SER) of 9.6×10^{-3} . In back-to-back (B2B) operation at 113-Gbaud PAM4, a TDECQ of 2.8 dB and an extinction ratio of 4.8 dB were obtained, corresponding to an aggregate line rate of 226 Gb/s.

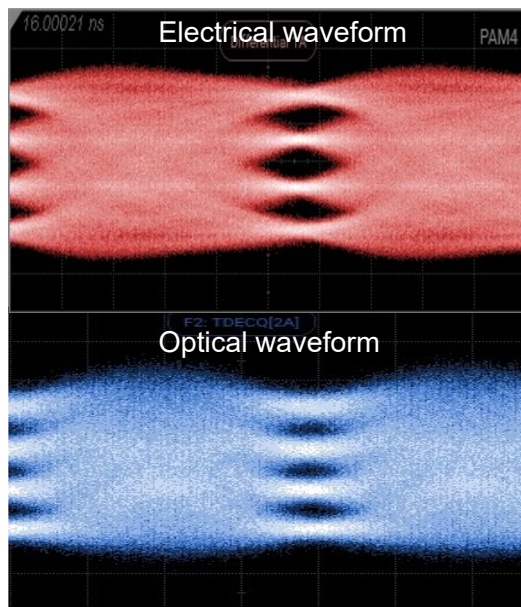


Fig. 6: Measured optical eye diagrams of 226 Gb/s (113GBaud PAM4) at back-to-back transmission

Conclusion

We have demonstrated a flip-chip-mounted differential EML with a dual-EAM structure on a glass interposer, with integrated 100-Ω differential terminations and backside AC-coupling

capacitors. The package-aware sub-assembly provides a short electrical path to each modulator section, resulting in a differential 3-dB EO bandwidth of 60.3 GHz and back-to-back (B2B) operation at 226 Gb/s (113-Gbaud PAM4) with a TDECQ of 2.8 dB. The proposed architecture directly addresses the passive-component footprint and high-frequency degradation issues that limit the density of differential transmitters, and is a promising approach for multichannel, high-density optical transmitters for next-generation AI and data-centre interconnects.

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